Navya S

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LinkedIn | GitHub | LeetCode | HackerRank

Objective

Computer Science undergraduate with experience in AI-powered applications,backend development and REST APIs. Passionate about building scalable software systems and intelligent applications using Python, FastAPI, and modern machine learning technologies.

Key Skills

Programming: Python, Java
Core CS: Data Structures & Algorithms, Object-Oriented Programming
AI/ML & LLM: RAG, Prompt Engineering, Semantic Search, Text Embeddings, FAISS, HuggingFace Transformers
Backend: FastAPI, REST APIs, API Integration, HTTP Methods
Databases: MySQL, PostgreSQL
Tools: Git, GitHub, Postman, VS Code
Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, OpenCV

Projects

User Management API GitHub

Tech Stack: Python, FastAPI, PostgreSQL, SQLAlchemy

- Designed and developed scalable RESTful APIs for user management using FastAPI.
- Implemented database integration with PostgreSQL and SQLAlchemy ORM.
- Ensured data validation and integrity using Pydantic models.
- Tested and debugged five API endpoints using Postman and Swagger UI.

Brain Tumor Classification GitHub

Tech Stack: Python, TensorFlow, OpenCV, Scikit-learn

- Developed a MobileNetV2-based CNN achieving 95% accuracy on a 4-class MRI dataset.
- Applied Grad-CAM for explainable AI, improving interpretability of model predictions.

Calorie Burnt Prediction API GitHub

Tech Stack: Python, FastAPI, Scikit-learn, XGBoost

- Built a backend system for calorie prediction using FastAPI and machine learning models.
- Designed API endpoints for real-time prediction based on user input.
- Integrated ML model with backend services for seamless inference.

Gen AI Study Assistant for Students GitHub

Tech Stack: Python, LangChain, FAISS, Hugging Face, Streamlit

- Built a document-based question-answering system using LangChain and FAISS.
- Designed a pipeline for processing PDFs and retrieving relevant information efficiently.
- Developed an interactive web interface using Streamlit for user interaction.

Internships

Lennox India Technology Center Pvt Ltd — Student Intern **2025**

- Tested and validated REST APIs, improving backend reliability.
- Debugged API response issues and enhanced system performance.
- Automated testing workflows using Selenium and Robot Framework.

CodSoft — Machine Learning Intern **2024**

- Built and evaluated machine learning models for classification tasks.
- Performed data preprocessing and feature engineering on 1000+ data samples to improve model performance.
- Developed end-to-end ML workflows ensuring efficient performance.

Education

KPR Institute of Engineering and Technology, Coimbatore **2023 – 2027**
B.E. Computer Science and Engineering (Artificial Intelligence & Machine Learning)
CGPA: 8.11/10

The TVS School, Madurai **2023**
HSC Percentage: 85.5%

Certifications

Introduction to Generative AI Concepts - Google Cloud	**2025**
Introduction to Machine Learning Concepts - Google Cloud	**2025**
Generative AI Concepts - Microsoft	**2025**
RDBMS PostgreSQL - IIT Bombay	**2025**
Python 3.4.3 - IIT Bombay	**2025**
Cyber Security and Privacy - NPTEL	**2025**
Introduction to Large Language Models(LLM) - NPTEL	**2026**